

Tokenized Option Ladder: Mathematical Formulation

1 Setup and the Splitting Primitive

1.1 Mechanism and Prerequisites

Maturity and clearing. A position is opened against collateral and settled at a single maturity m . At maturity an oracle posts the terminal underlying price X_m *once*; every tranche then clears to a closed-form cash (or collateral) amount. The oracle is therefore needed only at this clearing instant, not during the life of the position.

Two collateral states. Every construction starts from one of two collateral states: one unit of the underlying T (terminal value X_m), or cash C (in the put-side base split $C = S_0$).

Numeraire. Payoffs are written in cash / numeraire value by default; collateral-unit forms (underlying- T units, cash units) are given where useful.

Scope. This section is a payoff *decomposition*, not a pricing model. Premiums are set by the market; the math here only guarantees that the tranches partition the collateral's terminal value exactly.

1.2 The Splitting Primitive

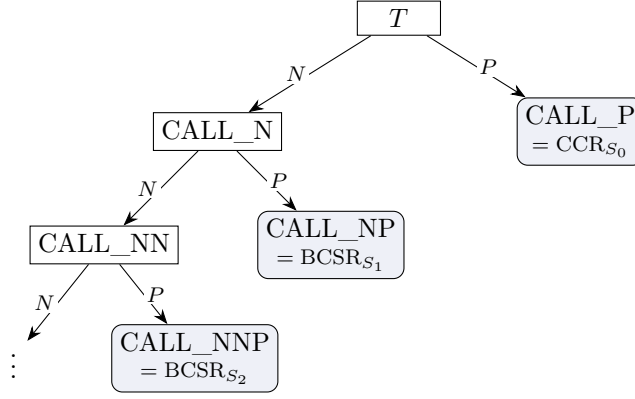
At a chosen strike S , any collateral position splits into an *obligation residual* P and a *right* N ,

$$\text{Collateral} = \underbrace{P}_{\text{residual}} + \underbrace{N}_{\text{right}}, \quad P(X_m) + N(X_m) = (\text{terminal value of the collateral}) \quad \forall X_m.$$

This conservation identity is what makes every tranche *fully collateralized*: the pieces always sum back to the collateral that backs them.

1.3 Tranche-Address Notation (the P/N Tree)

The suffix on each label is the path taken through the recursion. Starting from the collateral, every split sends the obligation residual to a leaf (P , a tranche the holder keeps) and keeps splitting the right (N):



The strike ladder *is* this recursion. Reading down the right edges gives the menu of tranches the protocol mints—CALL_P (the covered-call residual CCR_{S_0}), then CALL_NP, CALL_NNP, $\dots$ (the bull-spread residuals $BCSR_{S_1}, BCSR_{S_2}, \dots$)—while the left spine carries the remaining call right down to the tail call $CALL_{S_n}$. The running index i flattens the repeated- N path, so $BCSR_{S_i}$ is the residual produced at the i -th split.

1.4 Notation

Symbol	Meaning
X	Spot price of the underlying (running).
X_m	Terminal underlying price at maturity m , in cash/numeraire units; the one value the oracle posts at clearing.
T	One unit of underlying collateral; terminal cash value X_m .
C	Cash collateral; in the put-side base split $C = S_0$.
S	A strike price (generic).
S_i	Strike of the i -th split, $i = 0, \dots, n$; S_0 is the base strike. Call ladder: $S_0 < S_1 < \dots < S_n$; put ladder: $S_0 > S_1 > \dots > S_n$.
N	Right leg of a split: the long option right that is split off (+CALL or +PUT); the only leg that splits again, i.e. the recursing branch of the tree.
P	Obligation-residual leg of a split: the collateral the writer keeps after the right N is split off; it carries the short-option obligation. Every split obeys collateral = $P + N$.
$CALL_{S_i}$	Tokenized call position struck at S_i (a token, not a traded option); terminal value $\max(0, X_m - S_i)$.
PUT_{S_i}	Tokenized put position struck at S_i (a token, not a traded option); terminal value $\max(0, S_i - X_m)$.
$\pm CALL_{S_i}, \pm PUT_{S_i}$	Long (+) / short (−) the corresponding token.
CC	Covered call (strategy): hold the underlying and write a call against it, capping the upside above the strike in return for the premium.
CCR_{S_0} (CALL_P)	Covered-call residual token: the base-split residual leg $T - CALL_{S_0}$. Exists only at the base strike S_0 .

CSP	Cash-secured put (strategy): hold cash and write a put fully backed by that cash, taking on the obligation to buy the underlying below the strike.
CSPR_{S_0} (PUT_P)	Cash-secured-put residual token: the base-split residual leg $C - \text{PUT}_{S_0}$. Exists only at S_0 .
BCS	Bull call spread (strategy): long $\text{CALL}_{S_{i-1}}$, short CALL_{S_i} (the i -th rung), a bounded bullish payoff between the two strikes.
BCSR_{S_i} (CALL_NP)	Bull-call-spread residual token at the i -th split ($i = 1, \dots, n$): the residual leg of the bull call spread BCS, between strikes S_{i-1} and S_i .
BPS	Bear put spread (strategy): long $\text{PUT}_{S_{i-1}}$, short PUT_{S_i} (the i -th rung), a bounded bearish payoff between the two strikes.
BPSR_{S_i} (PUT_NP)	Bear-put-spread residual token at the i -th split ($i = 1, \dots, n$): the residual leg of the bear put spread BPS, between strikes S_{i-1} and S_i .

2 Route Strategy Map

StrikeX route strategy map

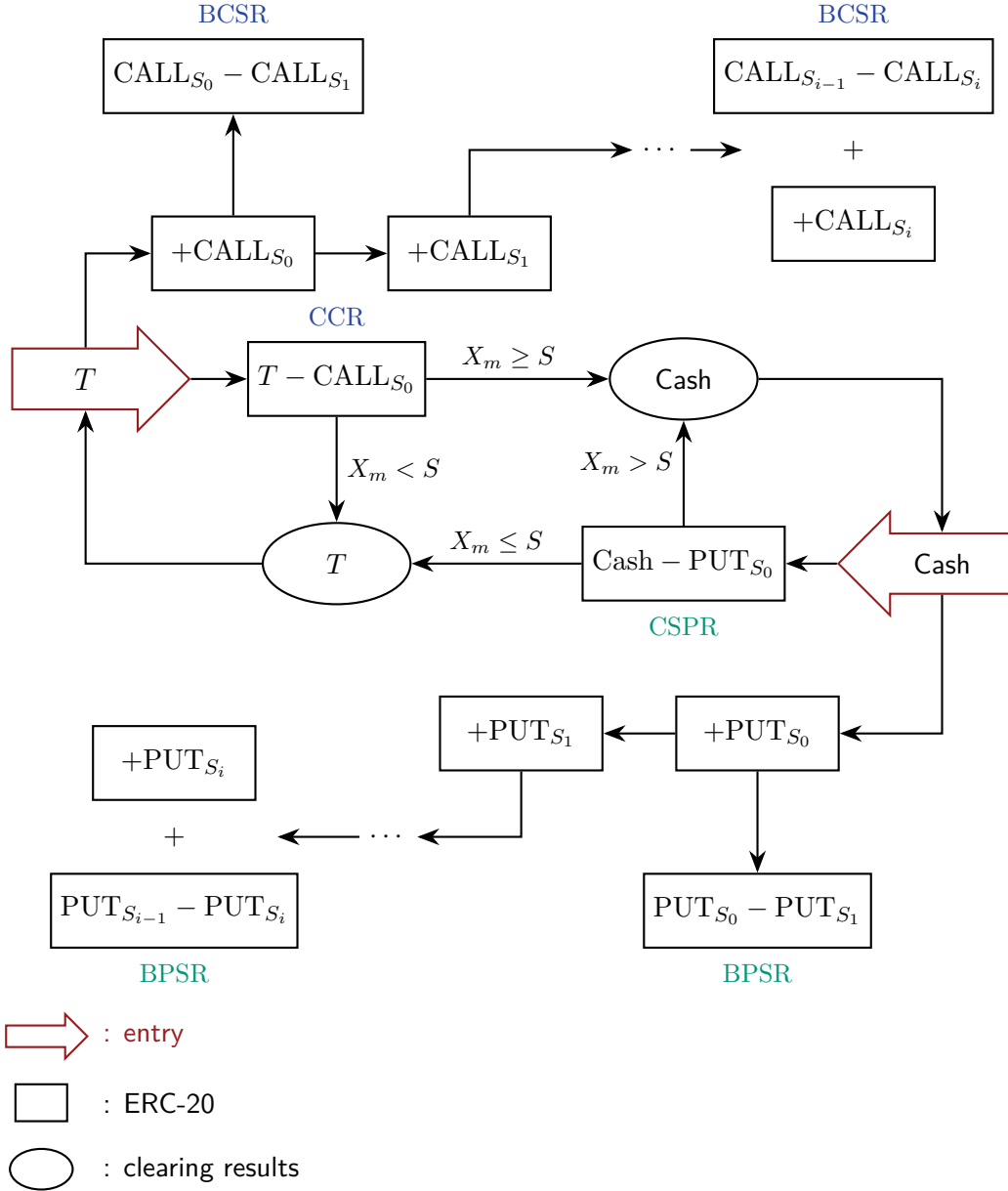


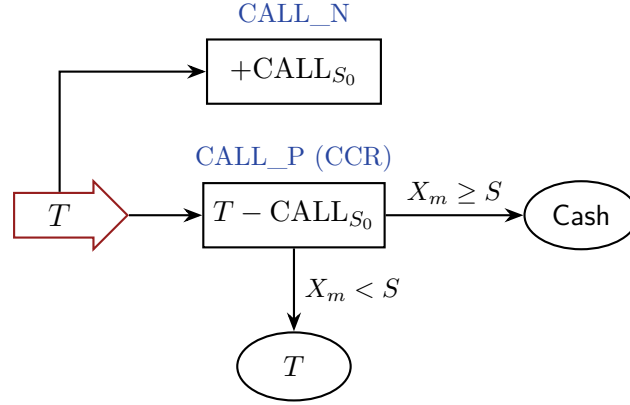
Figure 1: Route strategy map. From each collateral state, the base split yields an obligation residual (P , the $CCR_{S_0}/CSPR_{S_0}$ token) and a right (N); the residuals settle to cash or T at maturity according to X_m vs. the strike S , while the rights recurse along the strike ladder into the vertical-spread residuals BCSR/BPSR.

Figure 1 indexes the whole construction: each strategy of Sections 3–4 is one labeled route on this map, and Section 5 reads the same graph as a reinvestment cycle.

3 Long Underlying Assets

A long position in the underlying T (terminal value X_m) is tokenized into a ladder of call-side tranches. The base split (Section 3.1) carves T into a long-call token and a covered-call residual; the recursive split (Section 3.2) carves the long-call token into bull-call-spread tranches along the strike ladder; and the whole ladder conserves back to X_m (Section 3.3).

3.1 Call-Side Base Split & Covered Call



We start *long the underlying*: one unit of collateral T , worth X_m at maturity. At the base strike S_0 this long position is tokenized into two ERC-20 tranches—a right N and a residual P :

$$T = (N, P) \mapsto \text{CALL_N (long-call token)} + \text{CALL_P (covered-call residual)}.$$

(a) Splitting. At S_0 the underlying splits into the covered-call residual (the underlying minus the written call) and the call right:

$$T = (T - \text{CALL}_{S_0}) + \text{CALL}_{S_0}, \quad (1)$$

$$T = \text{CALL_P}_{S_0} + \text{CALL_N}_{S_0}. \quad (2)$$

$\text{CALL_P}_{S_0} = T - \text{CALL}_{S_0}$ is the covered-call leg—long the underlying, short the S_0 call, so its upside is capped at S_0 . $\text{CALL_N}_{S_0} = \text{CALL}_{S_0}$ is the long-call leg—the uncapped upside above S_0 , collateralized by the residual holder.

(b) Settlement. At maturity each token clears to a closed form:

$$\text{CALL_N}_{S_0}(X_m) = (X_m - S_0)^+ = \max(0, X_m - S_0), \quad (3)$$

$$\text{CALL_P}_{S_0}(X_m) = X_m - (X_m - S_0)^+ = \min(X_m, S_0). \quad (4)$$

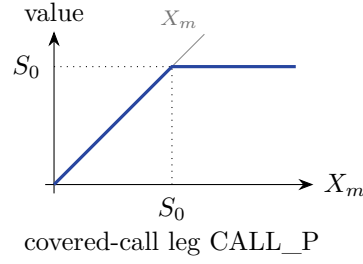
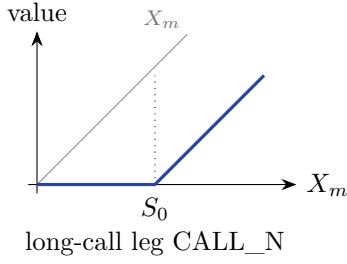
The covered-call leg converts the underlying into cash S_0 when the call is assigned ($X_m \geq S_0$) and keeps the underlying T , worth X_m , otherwise. The two legs conserve:

$$\text{CALL_P}_{S_0}(X_m) + \text{CALL_N}_{S_0}(X_m) = \min(X_m, S_0) + (X_m - S_0)^+ = X_m. \quad (5)$$

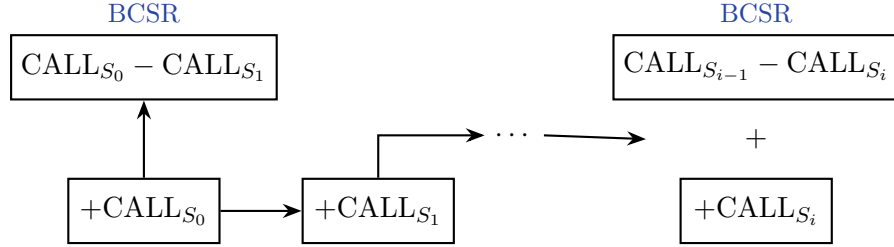
In underlying-collateral units ($X_m > 0$),

$$\text{CALL_P}_{S_0}^T = \min\left(1, \frac{S_0}{X_m}\right) T, \quad \text{CALL_N}_{S_0}^T = \max\left(0, 1 - \frac{S_0}{X_m}\right) T.$$

(c) Settlement charts. The two legs are the standard long-call and covered-call payoffs (USD settlement value vs. underlying price X_m):



3.2 Call-Side Recursive Split & Bull Call Spread



The long-call token CALL_N is itself split at the next rung. For a call ladder with increasing strikes $S_0 < S_1 < \dots < S_n$, each lower-strike call right splits into a capped bull-call-spread residual plus the higher-strike call carried further down the ladder.

(a) Splitting. For $i = 0, \dots, n-1$, the call right CALL_N_{S_i} is split at the next strike S_{i+1} :

$$\text{CALL_N}_{S_i} = (\text{CALL}_{S_i} - \text{CALL}_{S_{i+1}}) + \text{CALL}_{S_{i+1}}, \quad (6)$$

$$\text{CALL_N}_{S_i} = \text{CALL_NP}_{S_{i+1}} + \text{CALL_NN}_{S_{i+1}}. \quad (7)$$

$\text{CALL_NP}_{S_{i+1}}$ is the bull-call-spread residual (long the S_i call, short the S_{i+1} call); $\text{CALL_NN}_{S_{i+1}}$ is the higher-strike tail call — itself the call right $\text{CALL_N}_{S_{i+1}}$ handed to the next rung, which splits again.

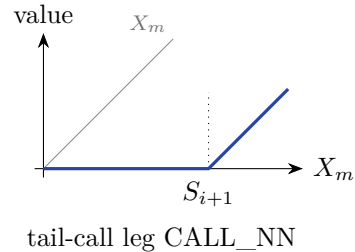
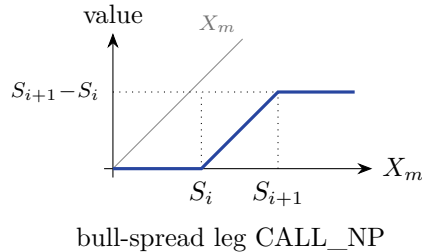
(b) Settlement.

$$\text{CALL_NP}_{S_{i+1}}(X_m) = (X_m - S_i)^+ - (X_m - S_{i+1})^+ = \min(\max(0, X_m - S_i), S_{i+1} - S_i), \quad (8)$$

$$\text{CALL_NN}_{S_{i+1}}(X_m) = (X_m - S_{i+1})^+ = \max(0, X_m - S_{i+1}). \quad (9)$$

Since $S_{i+1} - S_i > 0$, the spread residual is non-negative and capped at $S_{i+1} - S_i$.

(c) Settlement charts. The bull-spread leg is capped between adjacent strikes; the tail call is an ordinary long call struck at S_{i+1} :



3.3 Call-Side Conservation: Combined Chart and Table

After recursively splitting to S_n , the full call-side ladder identity is

$$X_m = \text{CCR}_{S_0}(X_m) + \sum_{i=1}^n \text{BCSR}_{S_i}(X_m) + \text{CALL}_{S_n}(X_m), \quad (10)$$

$$= \min(X_m, S_0) + \sum_{i=1}^n [(X_m - S_{i-1})^+ - (X_m - S_i)^+] + (X_m - S_n)^+, \quad (11)$$

$$= \min(X_m, S_0) + \sum_{i=1}^n \min(\max(0, X_m - S_{i-1}), S_i - S_{i-1}) + \max(0, X_m - S_n). \quad (12)$$

The right-leg sum is telescoping:

$$\sum_{i=1}^n [(X_m - S_{i-1})^+ - (X_m - S_i)^+] + (X_m - S_n)^+ = (X_m - S_0)^+, \quad (13)$$

so $\min(X_m, S_0) + (X_m - S_0)^+ = X_m$: the tranches always sum back to the underlying.

Numerical check. Parameters $S_0 = 2000$, $S_1 = 2500$, with

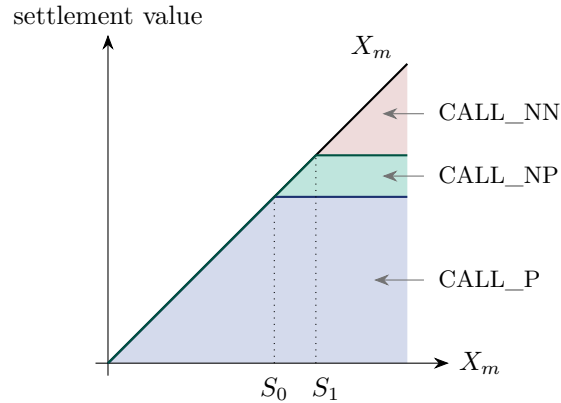
$$\text{CALL_P} = \min(X_m, 2000), \quad (14)$$

$$\text{CALL_NP} = \min(\max(0, X_m - 2000), 500), \quad (15)$$

$$\text{CALL_NN} = \max(0, X_m - 2500). \quad (16)$$

X_m	CALL_P	CALL_NP	CALL_NN	Total
3600	2000	500	1100	3600
2600	2000	500	100	2600
2500	2000	500	0	2500
2000	2000	0	0	2000
1700	1700	0	0	1700
1500	1500	0	0	1500
1000	1000	0	0	1000

Combined chart. Stacking the tokens at every price fills exactly up to the 45° line X_m — the visual statement of conservation ($S_0 = 2000$, $S_1 = 2500$):



4 Put Side

Begin with cash collateral $C = S_0$. At the base strike S_0 the cash splits into an obligation residual and a right:

$$C = (C - \text{PUT}_{S_0}) + \text{PUT}_{S_0}, \quad (17)$$

$$C = \text{PUT_P}_{S_0} + \text{PUT_N}_{S_0}. \quad (18)$$

The residual PUT_P_{S_0} is the cash-secured-put position (Section 4.2); the right PUT_N_{S_0} is the long put (Section 4.1), split further along the ladder into bear put spreads (Section 4.3). Unlike the call side, every put-side tranche conserves to the fixed cash value S_0 , not to X_m .

4.1 Long Put (PUT_N)

How it works. The long put is the *right* half of the cash base split: a tokenized put position struck at S_0 . Its holder owns the downside below S_0 and nothing above, giving short / bearish exposure to the underlying, while the matching residual (PUT_P) keeps the cash that collateralizes it. It is the only put-side leg that splits further.

Splitting. From the base split $C = \text{PUT_P}_{S_0} + \text{PUT_N}_{S_0}$, the long put is the right component

$$\text{PUT_N}_{S_0} = \text{PUT}_{S_0}. \quad (19)$$

Clearing (settlement).

$$\text{PUT_N}_{S_0}(X_m) = (S_0 - X_m)^+ = \max(0, S_0 - X_m). \quad (20)$$

In cash-collateral units, with $C = S_0$:

$$\text{PUT_N}_{S_0}^C(X_m) = \max\left(0, 1 - \frac{X_m}{S_0}\right) C. \quad (21)$$

[TODO: Split + clearing diagram for the long put (zoom-in of the wheel: the PUT_N right leg).]

[TODO: Settlement chart: $\text{PUT_N}_{S_0}(X_m)$ vs. underlying USD price X_m (45° down to S_0 , then flat at 0).]

[TODO: Settlement table for this tranche across X_m (the PUT_N column of the combined table in Section 4.4).]

4.2 Cash-Secured Put (PUT_P / CSPR_{S₀})

How it works. The cash-secured put is the *residual* half of the cash base split: the cash minus the put that was sold off. Its holder keeps the full cash S_0 when the put expires worthless ($X_m > S_0$) and is converted into the underlying (now worth X_m) when the put finishes in the money ($X_m \leq S_0$) — the cash-secured-put writer's position, used to acquire the underlying at a discount.

Splitting. From the base split,

$$\text{PUT_P}_{S_0} = C - \text{PUT}_{S_0}. \quad (22)$$

Clearing (settlement).

$$\text{PUT_P}_{S_0}(X_m) = \text{CSPR}_{S_0}(X_m) = S_0 - (S_0 - X_m)^+ = \min(X_m, S_0). \quad (23)$$

Settlement semantics: if $X_m \leq S_0$ the put is assigned and the residual converts the cash into the underlying T ; if $X_m > S_0$ it keeps the cash S_0 . In cash-collateral units, with $C = S_0$:

$$\text{PUT_P}_{S_0}^C(X_m) = \min\left(1, \frac{X_m}{S_0}\right) C. \quad (24)$$

Conservation with the long put:

$$\text{PUT_P}_{S_0}(X_m) + \text{PUT_N}_{S_0}(X_m) = \min(X_m, S_0) + (S_0 - X_m)^+ = S_0 = C. \quad (25)$$

*[**TODO:** Split + clearing diagram for the cash-secured put (the CSPR_{S_0} node: $\text{Cash} \rightarrow T$ when $X_m \leq S$, keep Cash when $X_m > S$).]*

*[**TODO:** Settlement chart: $\text{PUT_P}_{S_0}(X_m)$ vs. underlying USD price X_m (rising 45° to S_0 , then flat cap at S_0).]*

*[**TODO:** Settlement table for this tranche across X_m (the PUT_P column of the combined table in Section 4.4).]*

4.3 Bear Put Spread (PUT_NP / BPSR_{S_i})

How it works. Recursively split the put right at the next rung. For a put ladder with decreasing strikes

$$S_0 > S_1 > \dots > S_n,$$

each higher-strike put splits into a capped bear put spread residual plus the lower-strike put that is carried further down the ladder.

Splitting. For $i = 1, \dots, n$:

$$\text{PUT_N}_{S_{i-1}} = (\text{PUT}_{S_{i-1}} - \text{PUT}_{S_i}) + \text{PUT}_{S_i}, \quad (26)$$

$$\text{PUT_N}_{S_{i-1}} = \text{PUT_NP}_{S_i} + \text{PUT_NN}_{S_i}. \quad (27)$$

Clearing (settlement).

$$\text{PUT_NP}_{S_i}(X_m) = (S_{i-1} - X_m)^+ - (S_i - X_m)^+, \quad (28)$$

$$= \min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i), \quad (29)$$

$$\text{PUT_NN}_{S_i}(X_m) = (S_i - X_m)^+ = \max(0, S_i - X_m). \quad (30)$$

Sign convention check: $S_{i-1} - S_i > 0$ (put strikes decrease), so $\text{PUT_NP}_{S_i}(X_m) \geq 0$ and is capped at $S_{i-1} - S_i$.

*[**TODO:** Split + clearing diagram for the bear put spread (zoom-in: $\text{PUT_N} \rightarrow \text{PUT_NP} + \text{PUT_NN}$ at strike S_i).]*

*[**TODO:** Settlement chart: $\text{PUT_NP}_{S_i}(X_m)$ vs. underlying USD price X_m (flat cap $S_{i-1} - S_i$, ramp $S_i \rightarrow S_{i-1}$, flat 0).]*

*[**TODO:** Settlement table for this tranche across X_m (the PUT_NP column of the combined table in Section 4.4).]*

4.4 Put-Side Conservation: Combined Chart and Table

After recursively splitting to S_n , the full put-side ladder identity is

$$S_0 = \text{CSPR}_{S_0}(X_m) + \sum_{i=1}^n \text{BPSR}_{S_i}(X_m) + \text{PUT}_{S_n}(X_m), \quad (31)$$

$$= \min(X_m, S_0) + \sum_{i=1}^n [(S_{i-1} - X_m)^+ - (S_i - X_m)^+] + (S_n - X_m)^+, \quad (32)$$

$$= \min(X_m, S_0) + \sum_{i=1}^n \min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i) + \max(0, S_n - X_m). \quad (33)$$

The right-leg sum is telescoping:

$$\sum_{i=1}^n [(S_{i-1} - X_m)^+ - (S_i - X_m)^+] + (S_n - X_m)^+ = (S_0 - X_m)^+, \quad (34)$$

so $\min(X_m, S_0) + (S_0 - X_m)^+ = S_0$: the tranches always sum back to the cash collateral.

Numerical check. Parameters $C = S_0 = 2000$, $S_1 = 1500$, with

$$\text{PUT_P} = \min(X_m, 2000), \quad (35)$$

$$\text{PUT_NP} = \min(\max(0, 2000 - X_m), 500), \quad (36)$$

$$\text{PUT_NN} = \max(0, 1500 - X_m). \quad (37)$$

X_m	PUT_P	PUT_NP	PUT_NN	Total
3600	2000	0	0	2000
2600	2000	0	0	2000
2500	2000	0	0	2000
2000	2000	0	0	2000
1700	1700	300	0	2000
1500	1500	500	0	2000
1000	1000	500	500	2000
900	900	500	600	2000

[TODO: Combined put-side chart: stack PUT_P, PUT_NP, PUT_NN as USD settlement value vs. underlying price X_m , showing they sum to the flat line S_0 .]

5 Wheeling (Reinvest Strategy)

Figure 2 renders the construction of Sections 3–4 as a state machine over the two collateral states—the underlying T and cash C . Each red entry arrow splits its state into an obligation residual (CALL_P or PUT_P) and a right (CALL_N or PUT_N); recall X_m is the terminal underlying price and S the relevant strike. At maturity the covered-call residual CCR_{S_0} converts the underlying into cash when the call finishes in the money ($X_m \geq S$) and retains T otherwise ($X_m < S$); its mirror, the cash-secured-put

residual CSPR_{S_0} , converts cash into T when the put finishes in the money ($X_m \leq S$) and keeps cash otherwise ($X_m > S$). The right legs decompose recursively along the strike ladder into the bull and bear vertical-spread residuals BCSR and BPSR of Sections 3.2 and 4.3. This is the cycle that connects the two states across maturities, and it is where time enters: each clearing re-mints collateral into the opposite state. The closed forms for every quantity labeling the graph are collected after the figure.

Equations referenced in the diagram

Base splits (one maturity step):

$$T = \underbrace{(T - \text{CALL})}_{\text{CALL_P}} + \underbrace{\text{CALL}}_{\text{CALL_N}}, \quad \text{Cash} = \underbrace{(\text{Cash} - \text{PUT})}_{\text{PUT_P}} + \underbrace{\text{PUT}}_{\text{PUT_N}}.$$

Entry-split weights (token / cash units; here S denotes the base strike S_0 and X_m the terminal price):

$$\begin{aligned} \text{CALL_P} &= \min\left(1, \frac{S}{X_m}\right) T, & \text{CALL_N} &= \max\left(0, 1 - \frac{S}{X_m}\right) T, \\ \text{PUT_P} &= \min\left(1, \frac{X_m}{S}\right) C, & \text{PUT_N} &= \max\left(0, 1 - \frac{X_m}{S}\right) C. \end{aligned}$$

Per-bucket spread and tail payoffs:

$$\begin{aligned} \text{CALL_NP}_{S_i} &= \min(\max(0, X_m - S_{i-1}), S_i - S_{i-1}), & \text{CALL_NN}_{S_i} &= \max(0, X_m - S_i), \\ \text{PUT_NP}_{S_i} &= \min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i), & \text{PUT_NN}_{S_i} &= \max(0, S_i - X_m). \end{aligned}$$

6 Summary

Compact identities

Call side, increasing strikes $S_0 < S_1 < \dots < S_n$ (conserves to X_m):

$$X_m = \underbrace{\min(X_m, S_0)}_{\text{covered-call residual}} + \sum_{i=1}^n \underbrace{\min(\max(0, X_m - S_{i-1}), S_i - S_{i-1})}_{\text{bull call spread residual}} + \underbrace{\max(0, X_m - S_n)}_{\text{tail call}}. \quad (38)$$

Put side, decreasing strikes $S_0 > S_1 > \dots > S_n$ (conserves to S_0):

$$S_0 = \underbrace{\min(X_m, S_0)}_{\text{cash-secured-put residual}} + \sum_{i=1}^n \underbrace{\min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i)}_{\text{bear put spread residual}} + \underbrace{\max(0, S_n - X_m)}_{\text{tail put}}. \quad (39)$$

Master tranche reference

Label	Formula / meaning
CALL_P / CCR_{S_0}	Covered call residual: $\min(X_m, S_0)$.
CALL_N	Call right (long call): $\max(0, X_m - S_0)$.
CALL_NP / BCSR_{S_i}	Bull call spread residual: $\min(\max(0, X_m - S_{i-1}), S_i - S_{i-1})$.

CALL_NN	Higher-strike tail call: $\max(0, X_m - S_i)$.
PUT_P / CSPR $_{S_0}$	Cash-secured put residual: $\min(X_m, S_0)$.
PUT_N	Put right (long put): $\max(0, S_0 - X_m)$.
PUT_NP / BPSR $_{S_i}$	Bear put spread residual: $\min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i)$.
PUT_NN	Lower-strike tail put: $\max(0, S_i - X_m)$.

*[**TODO:** Master worked example combining both sides + the wheel cycle across two maturities (extends the per-side numerical checks of Sections 3.3 and 4.4).]*

StrikeX Wheeling strategy

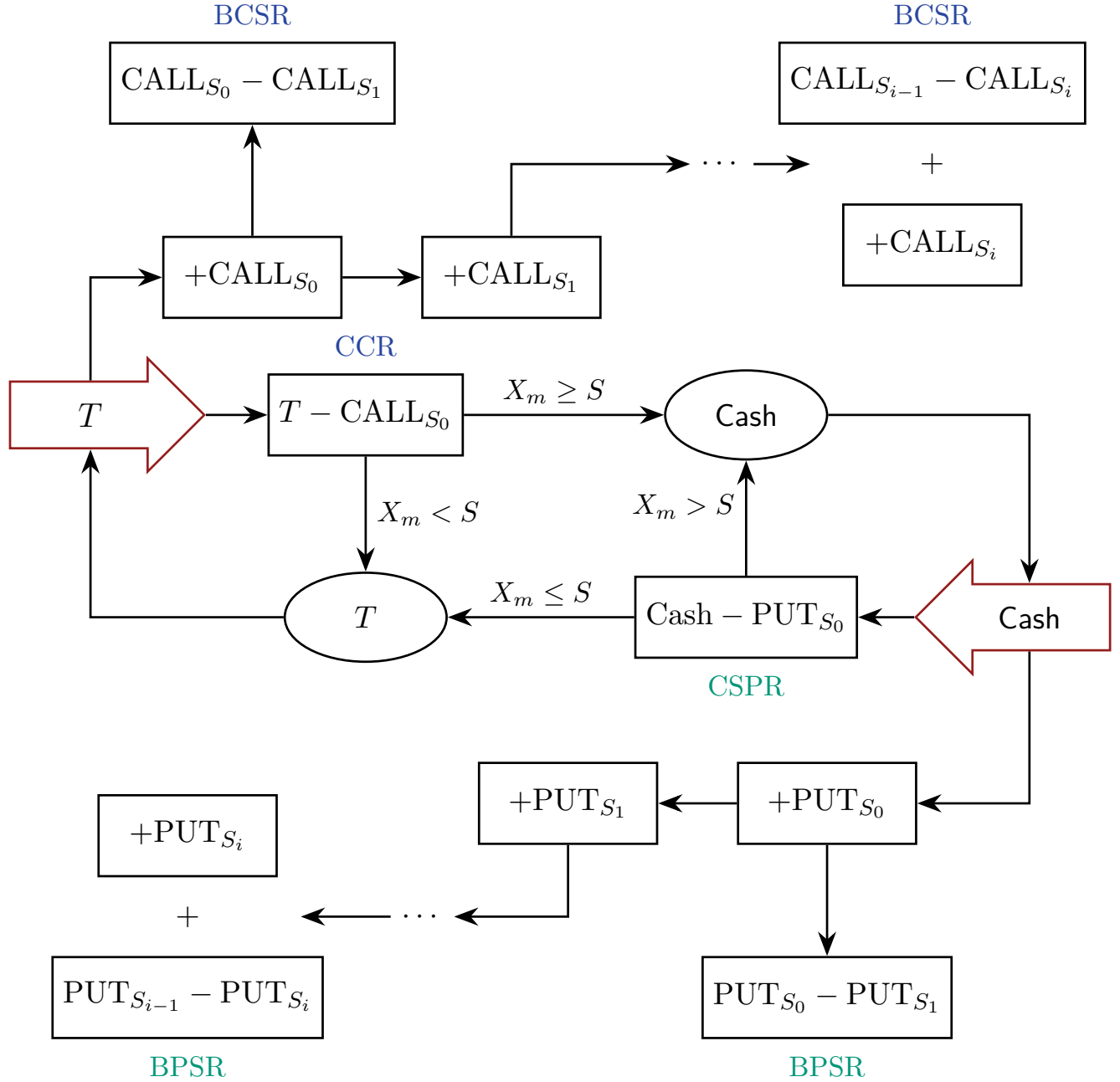


Figure 2: Tokenized option ladder as a recursion / state machine over the underlying state T and the cash state. Red arrows are entry points; each state splits into an obligation residual (P) and a right (N). The covered-call residual (CCR_{S_0}) and its mirror the cash-secured-put residual ($CSPR_{S_0}$) move probability mass between the T and $Cash$ states at maturity according to X_m vs. the strike S , while the right legs recurse along the strike ladder into bull/bear spread residuals (BSR, BPSR). This figure is the visual counterpart of the algebraic identities in Sections 3–4.